

Temporal Pixel Masking at 3.91-ms UVC Exposure: A Profile-Level Conventional-OCR Measurement Study

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[PLACEHOLDER: Funding acknowledgment to be filled.] This manuscript includes a planned minimal-risk human-subject study. No participant data have been collected. Recruitment will begin only after a written approval, exemption, or waiver determination by the competent institutional or departmental authority; the planned safeguards are described in Section V-F.

ABSTRACT Screen content can be silently captured and processed by optical character recognition (OCR) or vision-language models (VLMs). This study measures whether temporal display processing reduces conventional-OCR recovery at one fixed manual exposure on a single USB Video Class (UVC) camera link. The pipeline partitions each frame into rapidly alternating complementary subframes and adds profile-dependent overlays. We collected 10,575 physical captures across 9 distance–angle configurations. The comparable 3.91-ms subset contains 288 unprotected, 408 readability-priority, and 288 high-suppression short-exposure captures. Using the per-capture best result from Surya, EasyOCR, and Tesseract on geometrically corrected camera crops without adaptive contrast enhancement, mean character recovery was 94.5% unprotected, 16.7% for readability-priority, and 5.6% for high-suppression. The high-suppression profile therefore did not meet its strict $< 5\%$ target. Average recovery also hides failures: readability-priority short exposure has a pooled 94.5% P99 recovery, and pooled sensitive-token recovery is 24.0%. A 150-frame temporal mean and on-axis VLM probes recovered substantial text, placing sustained recording and modern recognizers outside the supported condition. The experiment does not identify an exposure interval, isolate temporal sparsity from luminance and overlay effects, or establish cross-device protection. Its contribution is a bounded empirical measurement: under this fixed UVC condition the evaluated profiles reduce conventional-OCR recovery, while the same archive identifies the integration, VLM, and usability-evidence boundaries that prevent a deployment or general screen-photography defense claim.

INDEX TERMS Adversarial perturbation, camera capture, display privacy, OCR, persistence of vision, screen-camera capture, temporal masking, visual eavesdropping

I. INTRODUCTION

SCREENS are ubiquitous in offices, banks, hospitals, and conference rooms, and so are capture devices: smartphones, smart glasses [1] (e.g., Ray-Ban Meta), and miniature cameras. A single photograph of a screen suffices to extract text via OCR or identify interface elements through object detection. This *visual eavesdropping* is covert: it leaves no network traces and requires no physical contact with the target device. Controlled experiments and field studies confirm that such attacks are common, succeed at high rates, and frequently go unnoticed [2], [3]. Although these devices motivate our work, the physical evidence presented here covers only a single UVC webcam; reproducibility on smartphones, smart glasses, and other image signal processor

(ISP) pipelines is left to future work.

A. EXISTING APPROACHES AND THEIR LIMITATIONS

Current screen-privacy solutions each occupy a limited niche. Spatial, software, watermarking, and temporal approaches address different viewing geometries or attack stages, but none directly evaluates on-axis short-exposure capture of static sensitive text followed by machine recognition; §II gives the detailed comparison. This leaves a concrete gap between restricting observation and reducing machine recovery from a physical snapshot.

The difficulty is that degrading information for the camera also tends to degrade it for the legitimate viewer. We ask whether display temporal structure can widen this gap under

short-exposure capture.

B. CORE IDEA

Our approach exploits the gap between human temporal integration and camera exposure windows. Each frame is decomposed into n randomly complementary subframes that alternate at high speed. When the exposure is shorter than or close to one subframe duration, the camera records only an incomplete pixel subset, whereas the human observer integrates content across subframes (Fig. 1). Because camera exposure is not instantaneous sampling, long exposure or video reconstruction covering a full cycle can re-sum the complementary subframes; we therefore make no irrecoverability guarantee for arbitrary exposure conditions.

We therefore frame this paper as a controlled measurement study rather than a deployment claim. It asks how much profile-level temporal display processing changes conventional OCR recovery from isolated captures at a fixed 3.91 ms UVC exposure, and where that observation fails. The concrete operational case is an instrumented UVC collection pipeline in which each target is represented by one fixed-exposure image and processed later by conventional OCR. It is not ordinary smartphone photography, smart-glasses capture, or a targeted operator who can keep recording, adjust exposure, apply adaptive enhancement, or invoke a VLM. Across the 8 geometries acquired at the common 3.91 ms setting, mean conventional-OCR character recovery was 94.5% unprotected, 16.7% for the designated readability-priority profile, and 5.6% for high-suppression. The high-suppression profile therefore did not meet its strict $< 5\%$ target. The defensible claim is this profile-level observation under one manual-exposure UVC condition, not an exposure interval, a general anti-camera setting, or a mechanism-isolated proof that temporal sparsity alone caused the reduction.

The boundary evidence is part of the measurement contribution, not an afterthought. Ambient illuminance was not recorded, and the logged UVC control settings cannot be ranked against typical smartphone auto-exposure. The evaluated temporal-mean attack averaged 150 short-exposure frames, nominally 2.5 s at 60 fps; shorter video durations were not evaluated. VLM probes recover substantial large-font short snippets, showing that conventional-OCR suppression does not transfer to modern end-to-end recognizers (§V-C).

C. CONTRIBUTIONS

The main contributions of this paper are:

- 1) **Controlled profile-level physical measurement:** Building on prior temporal masking ideas, we implement pixel-level complementary subframes with profile-dependent stripe/glyph overlays and a GPU rendering prototype. The archive contains 10,575 captures from one eMeet SmartCam S600 across 3 distances and 3 angles. In the comparable 3.91-ms subset, mean character recovery is 94.5% unprotected, 16.7% for the designated readability-priority profile, and 5.6% for high-suppression. This establishes a fixed-condition

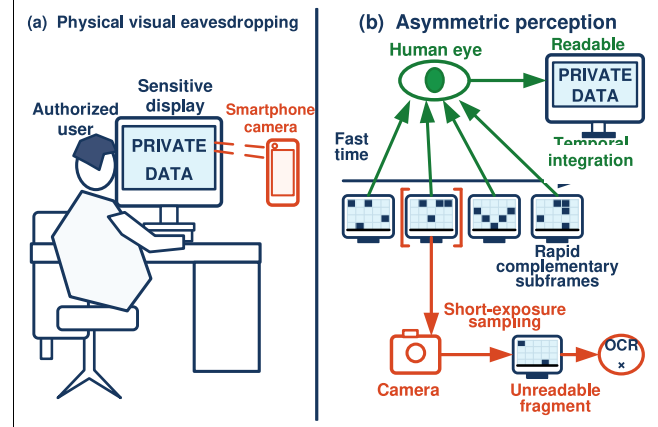


FIGURE 1. Threat scenario and operating boundary of temporal pixel masking. (a) A nearby camera can capture sensitive on-screen content without accessing the host system. (b) Human vision integrates complementary subframes, while a short camera exposure can intersect only part of a display cycle and leave OCR with an incomplete fragment. The camera path is conceptual: it does not assume zero-duration sampling or a fixed human integration time, and long exposure or multi-frame reconstruction can recover much more information.

observation for geometrically corrected camera crops processed by three open-source OCR engines without adaptive contrast enhancement.

- 2) **Cluster-aware descriptive evidence and exposure sensitivity:** We report capture-level distributions and content-cluster paired contrasts. Across all 9 archived geometries, unprotected short exposure exceeds readability-priority by 78.0 percentage points (content-cluster resampling interval [75.5, 80.3]) and exceeds high-suppression by 89.1 percentage points ([85.0, 92.4]). Restricting the analysis to the 8 common-setting geometries gives corresponding contrasts of 76.7 points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]). The intervals describe sensitivity to the 12 archived content clusters while treating the tested geometries as fixed; they do not quantify cross-device or cross-scenario uncertainty.
- 3) **Failure-boundary mapping:** Per-sample strongest digital attack recovery reaches 95.2%. A 150-frame camera mean recovers 71.1% for readability-priority and 47.9% for high-suppression, rising to 79.7% and 53.9% after the underexposed geometry is excluded. VLM probes recover substantial large-font content. These negative results define where the narrow conventional-OCR observation stops: readability-priority is the only profile tied to the planned usability protocol, whereas high-suppression remains an exploratory security-oriented stress profile without user-acceptability evidence.

The remainder of this paper is organized as follows. Section II reviews prior work, Section III defines the threat model, and Section IV presents the design. Section V reports the primary OCR evidence, VLM and integration boundaries, and the deferred usability protocol. Sections VI and VII syn-

thesize the evidence hierarchy and conclude the paper. Detection/tracking diagnostics are separated into supplementary material because they do not extend the primary OCR claim.

II. RELATED WORK

A. SCREEN VISUAL EAVESDROPPING THREATS

Remote screen reading attacks (telescopes, telephoto lenses) [4], deep-learning reconstruction of HDMI electromagnetic leakage [5], covert capture by smart glasses [1], and indirect eavesdropping through reflections and refractions [6] have all been documented. A controlled experiment by the Ponemon Institute found that 91% of visual hacking attempts succeed within 15 minutes [2]. Eiband et al. [3] surveyed 174 participants on the prevalence of shoulder surfing and the coping strategies users adopt. Tang and Shin [7] proposed Eye-Shield, which uses the front-facing camera to detect bystanders and blur sensitive regions in real time, though it relies on user-side hardware and provides only localized masking. Our work focuses on display-side mitigation that requires no attacker-side cooperation, without claiming prevention of information acquisition under all exposure and reconstruction conditions.

B. DISPLAY PRIVACY PROTECTION

Physical privacy filters restrict viewing angles but cannot block an on-axis camera. Chen et al. proposed HideScreen, a display-side shoulder-surfing defense that suppresses low-frequency visual cues so screen content blends with the background beyond a designed viewing distance [8]; it targets human shoulder surfing in the spatial domain rather than short-exposure camera capture or machine recognition. Zhu et al. [9] proposed LiShield, which modulates smart LED illumination to interfere with rolling-shutter cameras, but the scheme acts on ambient lighting rather than displayed content. Visual cryptography splits a secret image into shares that combine upon overlay [10], inspiring temporal decomposition ideas, though its original formulation targets printed media. Digital watermarking [11] can embed tracking identifiers, including screen-shooting-resilient variants [12], but provides post-hoc attribution rather than pre-emptive blocking. Software-based digital rights management (DRM) prevents digital screenshots but not physical photography.

Screen-camera visible light communication (VLC) [13]–[15] similarly exploits human temporal integration to embed information imperceptible to the eye yet decodable by cameras. Our method is VLC’s mirror image: VLC aims to let cameras *read* hidden information, whereas we aim to reduce machine readability of displayed content in constrained short-exposure capture.

Yerazunis and Carbone [16] first proposed temporal image masking for display privacy. Wu and Zhai [17] systematized temporal psychovisual modulation, and Gao et al. [18] applied it to information-security displays. Zhang et al. [19] proposed Kaleido, which re-encodes video frames via multi-frame decomposition to exploit the channel difference between display-eye and display-camera paths, making films

“viewable but not recordable.” Rainbow [20] addresses a related but distinct physical-world piracy setting: it modulates ambient illumination to corrupt rolling-shutter photographs of objects such as artwork, rather than temporally decomposing screen content. These schemes evaluate continuous recording or illumination attacks using imaging and subjective-quality metrics that are not interchangeable with the OCR, detection, and VLM recovery rates used here. None directly evaluate short-exposure single-frame capture of static screen text coupled with machine recognition. Mechanistically, Kaleido operates in continuous color space with chrominance-complementary frames and luminance variation, relying on perceptually complete mixing within a cycle; our method uses randomized discrete pixel-level partitioning, and the high-suppression profile deliberately abandons strict completeness to resist integration attacks (§IV-E). Lim [21] used persistence of vision to generate anti-screen-capture keypads in a browser, where a single screenshot captures only partial patterns. Table 1 situates the present study among representative spatial, illumination, and temporal schemes rather than treating any one baseline as a complete substitute.

The contribution is not temporal partitioning in isolation. It is a system-level measurement of a pixel-randomized implementation under a fixed UVC condition, together with explicit integration, sensitive-token, and VLM failure evidence. Detection, tracking, and digital usability proxies remain supplementary diagnostics.

C. ADVERSARIAL PERTURBATIONS AND PHYSICAL-WORLD ROBUSTNESS

Digital adversarial perturbations [22]–[24] and physical-world adversarial patches [25]–[27] are surveyed by Wei et al. [28]. Athalye et al. [29] achieved the first cross-viewpoint-robust 3D physical adversarial objects via Expectation over Transformation (EoT), though their approach still requires white-box access to the target model. A large digital-to-physical robustness gap persists: pixel-level digital perturbations are weakened considerably after camera sampling and JPEG compression [25]. Physical adversarial patches exhibit cross-viewpoint robustness but require perturbation patterns pre-generated for specific classifiers [26], [27], with limited transferability to unseen models [30]. The screen-camera link also introduces moiré artifacts, which Niu et al. [31] demonstrated can themselves act as adversarial perturbations. The base mechanism partitions content temporally at the display layer rather than optimizing a model-specific spatial perturbation. Enhanced profiles additionally add noise and stripe/glyph overlays, whose physical transfer is evaluated empirically rather than assumed. Physical photography is inherently a sampling process, and the defense is embedded in the display itself.

III. THREAT MODEL

A. PROTECTED ASSETS AND EVALUATED CONTENT

The motivating assets are sensitive on-screen text and visual information, including passwords, financial data, personally

TABLE 1. Scope Comparison with Representative Display and Physical-Scene Privacy Schemes. Entries Describe Different Threats and Evaluation Targets; They Are Not Matched-Parameter Performance Claims.

Scheme	Protected medium	Target threat	Mechanism	Reported evaluation
HideScreen [8]	Mobile display	Human shoulder surfing beyond a designed distance	Spatial suppression of low-frequency cues	Human-viewing distance/readability
LiShield [9]	Physical scene	Rolling-shutter camera capture	Modulated ambient LED illumination	Captured-image interference
Kaleido [19]	Displayed video	Continuous-recording piracy	Perceptually complete chrominance/luminance re-encoding	Recorded-video signal-to-noise ratio and subjective quality
Rainbow [20]	Physical objects	Mobile-camera piracy	Ambient-illumination modulation	Rolling-shutter image degradation
Lim [21]	Browser keypad	Single digital screen capture	Persistence-of-vision partial patterns	Keypad capture demonstration
This work	Static screen text	Short-exposure physical capture followed by machine recognition	Randomized pixel partitioning; high-suppression profile breaks completeness	OCR/VLM recovery and integration boundaries

identifiable information (PII), source code, confidential documents, and similar content. The present experiments should not be read as showing that all of these assets are protected by the readability-priority profile. Instead, the corpus treats credentials, digit strings, URLs/paths, code, paragraphs, and document pages as evaluated content types, with sensitive tokens reported separately because short high-value fields are harder to protect than dense text.

B. ATTACKER CAPABILITIES

Attack capabilities do not form a strict monotonic hierarchy but are jointly determined by exposure, geometry, temporal sampling, and post-processing capabilities. Results are reported across the following scope:

- **Primary scope:** Instrumented snapshots at the logged 3.91 ms UVC exposure setting, followed by per-frame recognition with Surya, EasyOCR, and Tesseract. The attacker may run all three engines but the evaluated pipeline does not include adaptive contrast enhancement or observations covering a full display cycle. This models isolated fixed-exposure UVC images, not ordinary auto-exposure phone snapshots or a targeted operator who can optimize capture and preprocessing. Ambient illuminance and display-camera phase were not recorded. This is a narrow measured condition, not the strongest rational attacker.
- **Practical failure boundary:** Long exposure, video temporal averaging after continuous recording, off-axis capture, phase search and channel reconstruction given a known period, object detection/association tasks, and end-to-end recognition by commercial vision-language models (§V-C). The evaluated video attack averages 150 short-exposure frames, nominally 2.5 s at 60 fps. It demonstrates a practical sustained-recording bypass but does not establish recovery from an arbitrary four-frame burst. We quantify these risks but interpret high-suppression results only as mitigation, not as a blocking guarantee.

- **Not yet adequately covered:** Additional camera/display combinations, precise rolling-shutter row alignment, real-world temporal super-resolution, learning-based reconstructors trained on display sequences, and long-term adaptive attacks.

C. ATTACKER KNOWLEDGE

The attacker *knows* the temporal masking algorithm, the sub-frame count n , and candidate period parameters, but does *not know* the current cycle's mask seed or phase starting point. The cryptographically secure pseudorandom number generator (CSPRNG) provides per-cycle mask randomization to prevent fixed-pattern learning and cross-cycle correlation; it does not provide a cryptographic guarantee against full-cycle averaging attacks. Once an attacker obtains and registers a complete cycle, an unknown seed cannot prevent linear reconstruction. More fundamentally, the method's security against short-exposure capture rests on disrupting the binarization and segmentation stages of conventional OCR pipelines, not on an information-theoretic impossibility of recovery from sparse pixels. End-to-end recognizers such as VLMs bypass binarization via visual encoding and language priors, and can recover substantial text from the same sparse fragments (§V-C).

D. PROTECTION GOALS

We use tiered design goals under short exposure to interpret the evaluated profiles. High-suppression targets conventional OCR character recovery $< 5\%$ with exact match $= 0\%$, whereas readability-priority targets a relative reduction of at least 80% with exact match $< 1\%$. These manuscript-level thresholds were not preregistered security guarantees. They do not apply to other exposures, adaptive preprocessing, VLMs, or other devices. Changes in object-detection mean average precision at 0.50 intersection over union (mAP50) are supplementary diagnostics rather than pass criteria. The ΔE_{00} and structural similarity (SSIM) values are digital reconstruction proxies, not evidence of real-panel perception or

user comfort.

IV. SYSTEM DESIGN

A. TEMPORAL SUBFRAME DECOMPOSITION

Given a normalized original frame $\mathbf{I} \in [0, 1]^{H \times W \times C}$, we generate binary masks $\mathbf{M}_1, \dots, \mathbf{M}_n$ satisfying $\sum_{k=1}^n \mathbf{M}_k = \mathbf{1}$, and set $\mathbf{I}_k = \mathbf{I} \odot \mathbf{M}_k$. This yields:

- **Digital-domain completeness:** $\sum_{k=1}^n \mathbf{I}_k = \mathbf{I}$;
- **Mutual exclusivity:** each pixel is illuminated in exactly one of the n time slots.

A distinction between “summation” and “temporal averaging” is necessary. On an ideal linear display with equal slot durations and equal peak luminance, the average radiance over one cycle is $\mathbf{I}/n$, not $\mathbf{I}$. Digital-domain completeness means subframes can be re-summed; it does not imply that real display luminance has been losslessly preserved.

This distinction is also an experimental limitation. The physical capture study did not include a luminance-matched static dimming control, a matched noise-off version of the enhanced profiles, or photometric panel timing measurements. The observed profile-level reductions therefore cannot be attributed to temporal sparsity alone; duty-cycle luminance, static stripe/glyph overlays, panel response, camera exposure, and ISP processing remain entangled.

Pixel assignments are generated from a ChaCha20 [32] CSPRNG. Rejection sampling maps the byte stream to unbiased slot indices; a separate Fisher–Yates shuffle [33] randomizes subframe playback order. A chi-square slot-count check ($\alpha = 0.01$, $\text{df} = n-1 = 3$) is used only as an implementation sanity check, with at most 5 resampling attempts. It is not evidence of cryptographic strength. Randomization reduces fixed patterns and cross-cycle correlation but cannot prevent linear reconstruction of a registered full cycle. Fig. 2 illustrates the complete pipeline.

B. COMPLEMENTARY ADVERSARIAL NOISE

We overlay adversarial noise $\mathbf{N}_1, \dots, \mathbf{N}_n$ on the subframes, satisfying:

$$\sum_{k=1}^n \mathbf{N}_k = \mathbf{0} \quad (1)$$

Equation (1) holds strictly only in the ideal linear domain prior to summation, clipping, and the display transfer function. Prior work has shown that small text-image perturbations can significantly mislead OCR systems [34], providing a basis for OCR-oriented noise design. Noise directions are generated by fast gradient sign method (FGSM)-style [22] proxy gradients with an amplitude bound of $\varepsilon = 8/255$, then decomposed complementarily in the temporal domain. Because displays cannot emit negative radiance, the implementation uses a pedestal level and numerical clipping. Consequently, after gamma correction, quantization, clipping, and camera imaging, the noise is not guaranteed to cancel physically.

Component role: The random dot-pattern mask is the primary temporal mechanism. Complementary noise is an

optional ideal-domain component; stripe/glyph overlays and partial inversion are nonlinear additions used by the enhanced profiles below. The archived composite-profile definitions are retained without retroactively removing components. This preserves correspondence with the captured data but does not establish component benefit or optimality. In the physical results, noise is nonmonotonic, and matched noise-off versions of the enhanced profiles were not captured. We therefore interpret complete profiles rather than recommend complementary noise as a default component.

C. LUMINANCE COMPENSATION

Each pixel is illuminated only once per n -slot cycle. If the display could boost light output by a factor of n in the linear-light domain during the active slot, the cycle-averaged luminance would approach the original. The current proof of concept (PoC) implements neither per-slot dynamic backlight coordination nor photometric verification. All luminance alignment, CIEDE2000 ΔE_{00} [35], and SSIM [36] results therefore come from digital integration/normalization models and serve only for cross-configuration comparison, not as evidence of real-panel optical equivalence or user comfort.

D. TIMING AND BANDWIDTH

For a basic cycle containing only n complementary subframes, taking 60 Hz as the minimum cycle rate requires $f_r \geq 60n$; at $n = 4$ this corresponds to 240 Hz. If the cycle additionally includes one inversion frame, the requirement becomes $f_r \geq 60(n + 1)$, and at 240 Hz the actual cycle rate drops to only 48 Hz. Thus, 240 Hz meets the basic four-subframe configuration but not the inversion-frame configuration’s 60 Hz cycle target. Human flicker perception also depends on luminance, contrast, retinal eccentricity, spatial content, and eye movements [37], [38]; nominal cycle rate alone cannot predict comfort.

Whether an observer integrates complementary subframes into a perceptually complete image depends on the temporal contrast sensitivity function (TCSF). Critical flicker fusion (CFF) is not a universal cutoff: Davis et al. [38] found sensitivity near conventional rates for uniform modulation but visibility above 500 Hz when high-frequency spatial edges were present. The 48 Hz full-cycle rate of the inversion configuration may therefore produce visible flicker for some observers, and the 60 Hz basic configuration cannot be assumed flicker-free. Only the planned user study can determine whether either configuration supports comfortable temporal integration.

E. EXPLORATORY HIGH-SUPPRESSION PROFILE

The basic linear temporal mask leaks under full-cycle temporal averaging (see §V-E2). To address this, we introduce nonlinear enhanced profiles. Unless otherwise stated, profiles in this subsection use $n = 4$ base temporal masking with complementary adversarial noise enabled. Table 2 summarizes all profile configurations; the “Mask + noise” row in Table 5 is the base condition without stripe/glyph anti-OCR overlay.

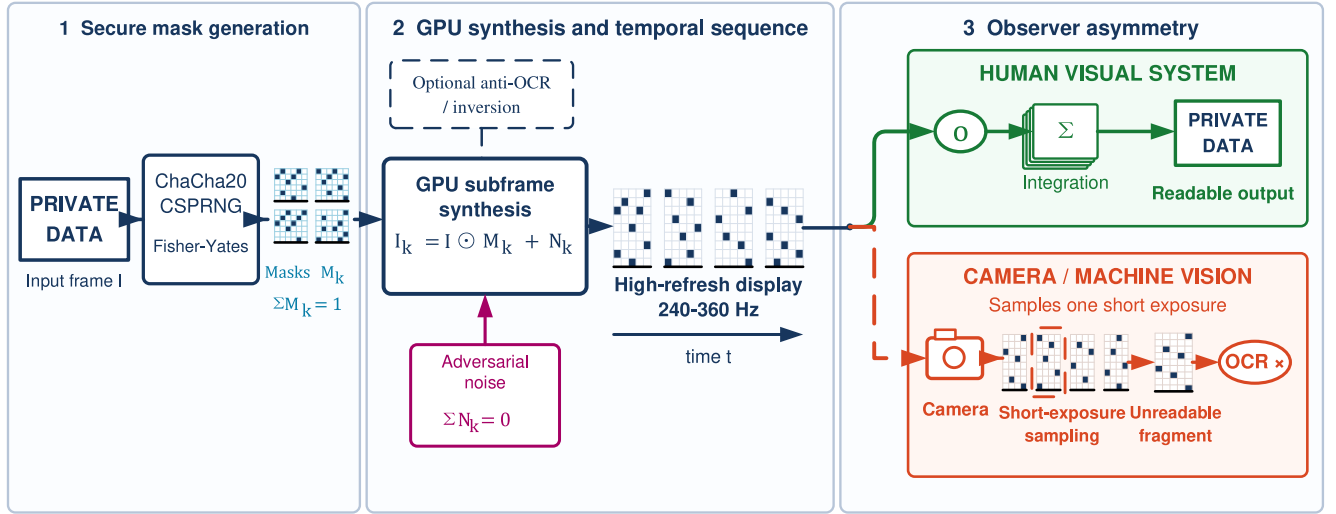


FIGURE 2. Temporal pixel masking system pipeline from source content to protected display. The input frame is decomposed into CSPRNG-assigned complementary pixel masks; optional complementary OCR-oriented noise, profile-dependent stripe/glyph overlays, and an optional partial inversion frame are then inserted before high-refresh playback. A human observer is expected to integrate multiple slots into a readable view, whereas a short-exposure camera may capture only a sparse subframe. Long exposure and video temporal averaging can accumulate multiple slots and are therefore evaluated separately as boundary attacks rather than assumed away.

TABLE 2. Profile Composition Summary ($n = 4$ for All Profiles)

Profile	Mask	Noise	Cell (px)	Stripe width (px)	Stripe α_s	Glyph α_g	Inv. α
Mask only	✓	–	1	–	–	–	–
Mask + noise	✓	✓	1	–	–	–	–
Strong	✓	✓	1	10	0.10	0.12	–
Readability-priority	✓	✓	1	10	0.10	0.12	0.2
High-suppression	✓	✓	2	6	0.42	0.55	0.2

- **Strong (anti-OCR) profile:** base mask+noise with mask cell size of 1 pixel, stripe width of 10 pixels, stripe amplitude $\alpha_s = 0.10$, and glyph amplitude $\alpha_g = 0.12$, without an inversion frame.
- **Readability-priority profile:** the Strong profile plus an $\alpha = 0.2$ inversion frame, designated in this paper as the usability-facing composite configuration. Archived result tables and capture metadata use the label “deployed,” but that legacy identifier is neither a deployment recommendation nor evidence of empirical optimization.
- **High-suppression profile:** an exploratory security-oriented stress profile using base mask+noise with mask cell size increased to 2 pixels, stripe width of 6 pixels, stripe and glyph amplitudes of 0.42 and 0.55 respectively, plus an $\alpha = 0.2$ inversion frame.

The nonlinear profiles depart from strict completeness ($\sum I_k \neq I$), producing visible stripes and grain. This trade-off is the opposite of Kaleido-class temporal re-encoding: the latter relies on perceptually complete mixing within a cycle to preserve viewing experience, meaning full-cycle integration inherently reconstructs an approximation of the original, consistent with its anti-piracy goal. In the static-content capture threat addressed here, completeness is precisely the entry

point for integration attacks; the high-suppression profile therefore makes breaking strict completeness an active design trade-off, accepting visible stripes and grain in exchange for lower recovery under full-cycle integration. Its security benefit and unmeasured readability cost must be reported separately, and it should not be read as a user-validated deployment candidate.

We use the paper-facing name “high-suppression” because it describes the profile’s relative position in Table 2 without implying that a capture-resistance target has been met or that users can tolerate the profile. The archived identifier remains `capture_hardened` (with legacy tag `v1m`) for reproducibility; §V-B reports the target miss and residual leakage.

F. PARTIAL INVERSION FRAME

A long-exposure attack occurs when the exposure window covers a full display cycle ($n = 4$ at 240 Hz yields a basic cycle of approximately 16.7 ms; with an inversion frame the $n + 1$ slot cycle is approximately 20.8 ms; §V-A reports measured long-exposure times of 31.25 and 125 ms). Such attacks approximate the original image through sensor integration. A direct countermeasure inserts a **partial inversion frame** $\alpha \cdot (255 - I)$ after each n -subframe cycle, introducing

TABLE 3. Inversion Frame Strength α Ablation on the Fixed Strong Overlay ($\alpha_s = 0.10$, $\alpha_g = 0.12$): Real-Capture Best-of-Engine Character Recovery (Mean and 95% Bootstrap CI, Five Account/Code/CET6 Items Pooled Across 9 Geometric Conditions; $N = 135$ per Setting for Long Exposure, $N = 45$ for Video Temporal Averaging). This Five-Item Subset Differs from the 12-Item Readability-Priority Rows in Table 5.

α	Long-exp. char (%)	Video avg. char (%)
0.0	68.6 [63.1, 73.9]	72.7 [63.3, 81.1]
0.2	67.0 [61.3, 72.7]	69.8 [59.9, 78.7]
0.3	72.0 [66.2, 76.9]	64.3 [54.0, 73.6]
0.5	61.7 [55.6, 67.7]	66.4 [56.5, 75.2]
1.0	18.1 [14.0, 22.0]	28.4 [20.5, 36.9]

a luminance component complementary to the original into the long-exposure integral and reducing net signal contrast. Higher α may reduce long-exposure recovery, but also alters the luminance and contrast of the digital integrated view.

A real-capture inversion strength ablation (Table 3; pipeline per §V-B) shows that weak inversion does not meaningfully reduce long-exposure recovery: $\alpha = 0$ yields 68.6% and $\alpha = 0.2$ yields 67.0% (best-of-engine), with 95% confidence intervals overlapping across $\alpha \in [0, 0.5]$. The ablation holds the Strong overlay fixed ($\alpha_s = 0.10$, $\alpha_g = 0.12$) and varies only inversion strength, using the five-item account/code/CET6 subset rather than the 12-item main corpus; its $\alpha = 0.2$ value ($N = 135$) is therefore not a repeated estimate of the readability-priority row's 60.9% ($N = 324$). Only near-full inversion ($\alpha = 1.0$) reduces recovery to 18.1%, at the cost of severely degrading the digital integrated view. Under the 150-frame video temporal mean, weak inversion similarly maintains 64–73% recovery. Weak inversion alone cannot resist integration attacks. The readability-priority choice of $\alpha = 0.2$ is a modeled design decision, not a user-study-validated optimum.

Each display cycle thus contains $n + 1$ time slots. At $n = 4$ and 240 Hz the cycle rate is 48 Hz, below the 60 Hz design target and potentially producing perceptible flicker; no fixed “flicker-free threshold” applies universally across viewing conditions.

G. IMPLEMENTATION SCOPE

The current PoC operates at the application/algorithm layer, using Python and GPU rendering to generate display sequences. It does not include OS driver-level injection (DXGI / KMS-DRM / CoreDisplay) or per-slot dynamic backlight control. We therefore validate the display sequence and its captured results, without claiming production-level system integration.

V. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

Hardware platform: Experiments ran on a desktop workstation with an Intel Core i7-8700K (3.70 GHz), 32 GB RAM, and an NVIDIA TITAN Xp (12 GB VRAM). The display is an AOC 24G51Z (23.8-inch Fast IPS panel, quantum-dot (QD) backlight, edge-lit LED without local dimming, RGB

stripe subpixels), driven at 1920×1080 with a nominal refresh rate of 240 Hz. The manufacturer specifies 1 ms gray-to-gray (GtG) response time; at 240 Hz each subframe slot lasts approximately 4.17 ms, so the GtG transition occupies roughly 24% of one slot. Actual gray-level transitions may differ from the nominal GtG specification, causing partial temporal overlap (ghosting) between adjacent subframes. The measured real-capture results already include these panel-response effects; mechanistic narrative should therefore be read as “camera samples a short exposure that may span partial transitions across subframes” rather than “camera captures exactly one discrete subframe.” This refresh rate satisfies only the basic $n = 4$ subframe cycle at $f_r/n = 60$ Hz. With an additional inversion frame per cycle, the actual cycle frequency is $240/(n+1) = 48$ Hz. Thus, the visual comfort of the inversion configuration cannot be inferred from “240 Hz high refresh rate” alone and still requires user-study validation.

Capture device: All physical captures use a single eMeet SmartCam S600 USB webcam with its standard UVC driver. The playback canvas was 1920×1080 ; camera output underwent perspective correction and content cropping before recognition. Short-exposure and video-frame settings were 0.49 or 3.91 ms, long-exposure settings were 31.25 or 125 ms, and video was captured at 60 fps. These are discrete DirectShow/UVC log₂ control values recorded in the capture metadata: -11 and -8 for short/video frames, and -5 and -3 for long exposure. Eight geometries use the common 3.91/31.25 ms settings; the early d0.5/15° run uses 0.49/125 ms. The term “fixed exposure” refers to these logged manual control settings, not to photometric calibration or representative smartphone shutter times. OCR inputs are geometrically corrected content crops. The primary OCR evaluation does not include an adaptive contrast-enhancement or preprocessing sweep.

The camera was placed at 0.5 m, 1 m, and 1.5 m from the display, rotated to 0°, 15°, and 30° at each distance, yielding 9 geometric configurations. Ambient illuminance was not recorded. The manufacturer specifies an 8-megapixel 4K image sensor and 1080p/60 fps output [39], but does not identify a Sony sensor; we therefore make no sensor-model claim. The 60-fps frame period is 16.67 ms, but rolling-shutter readout time and phase relative to the 4.17 ms display slots were not measured. Actual panel output and application-level slot timing were not verified photometrically. Row-level mixing and panel transitions may help explain why real-capture short-exposure recovery (15.1%) exceeds simulated single-subframe recovery (0–2.6%), but the current data do not isolate either mechanism. Results apply to this UVC link and should not be extrapolated to smartphones, smart glasses, or other camera ISPs.

Test content: At each of the 9 geometric positions, the real-capture OCR experiment used a fixed set of 12 content items: 11 selected synthetic items and one CET6 (College English Test Band 6) document (portrait 900×1280). Three captures per item give 36 captures per profile-attack con-

dition at each position and $12 \times 3 \times 9 = 324$ across all positions before the disclosed readability-priority-profile repeats. By contrast, the software-simulated OCR experiment used the complete 120-sample synthetic corpus (12 content templates $\times$ 10 variants, mapped to 9 category labels covering passwords, credentials, code, tables, paragraphs, URLs, etc.). Real-capture short-exposure OCR is the primary feasibility evidence. Detection/tracking diagnostics use separate COCO and MOT subsets and are reported only in the supplementary material because their metrics and experimental design do not carry the central OCR claim.

Evaluation metrics: OCR character recovery is $\max(0, 1 - \text{CER})$, where CER is character error rate. Exact match uses case-insensitive full-text comparison after whitespace removal. Sensitive-token recovery is the proportion of reference digits, account numbers, URL/path strings, and key-like fields recovered in full. Leak rate is the proportion of samples with character recovery $\geq 20\%$, an arbitrary descriptive threshold that does not imply that 19% recovery is safe. Distributional summaries are essential because the means hide a heavy upper tail. For pooled readability-priority short exposure, P50, P75, P90, P95, and P99 are 8.3%, 16.7%, 43.0%, 60.9%, and 94.5% ($N=459$). The corresponding high-suppression values are 1.6%, 8.3%, 14.1%, 18.7%, and 22.2% ($N=324$). These tails show that a small subset of close-range, large-font captures remains highly recoverable.

OCR aggregation and uncertainty: For each capture, each metric is maximized independently across the three engines. This metric-wise oracle models an attacker who retains whichever engine output is most favorable for character, exact, or token recovery. Table 5 reports descriptive 95% percentile-bootstrap intervals from 2,000 capture-level resamples with seed 20260612. They summarize the archived corpus and setup but are not population intervals because content and geometry are repeated. Paired contrasts match attack mode, content item, position, and available repeat index across profiles, average duplicate readability-priority rows before pairing, and resample the 12 content items as clusters. The resulting intervals characterize content-cluster sensitivity with the 9 tested geometries held fixed; they do not estimate uncertainty over new devices or geometries. Across all geometries, unprotected short exposure exceeds readability-priority by 78.0 percentage points (content-cluster resampling interval [75.5, 80.3]) and high-suppression by 89.1 points ([85.0, 92.4]) across 324 matched units. Mask-only exceeds readability-priority by 3.0 points ([-0.5, 6.5]), so the archive does not establish an independent benefit from every readability-priority component.

Common-setting analysis: The d0.5/15° position used a different 0.49-ms setting and produced visibly underexposed protected frames. The 8 geometries acquired at the common 3.91-ms setting are therefore the most internally comparable subset and define the primary fixed-setting result (Table 4). Their short-exposure means are 94.5% unprotected ($N = 288$), 16.7% readability-priority ($N = 408$), and 5.6% high-suppression ($N = 288$). The paired contrasts are

76.7 percentage points ([74.3, 79.2]) for unprotected minus readability-priority and 88.9 points ([85.5, 91.9]) for unprotected minus high-suppression. Excluding the noncomparable geometry also raises readability-priority 150-frame video-mean recovery from 71.1% to 79.7% [75.4, 83.8]% and high-suppression from 47.9% to 53.9% [48.0, 59.9]%. The full 9-geometry pool remains in Table 5 as a disclosed sensitivity analysis rather than the primary fixed-setting estimate.

Additional robustness checks: A leave-one-round-out analysis on the five repeated content items gives readability-priority short-exposure recovery of 16.6% for round 1 and 15.8% for round 2, leaving the profile-level interpretation unchanged. Detection and tracking differences have not undergone significance testing and are reported descriptively in the supplementary material.

Digital reconstruction proxies: ΔE_{00} is the mean CIEDE2000 color difference between the original image and the full-cycle digitally reconstructed image; SSIM [36] is their mean structural similarity. Both are computed in the digital pipeline and are used only for within-paper profile comparison. Neither metric measures real-panel optical output, temporal comfort, or human readability.

B. REAL-CAPTURE OCR EXPERIMENT

The physical archive comprises 10,575 images (1,175 per geometric position), all from the S600 webcam. This total mixes component ablations, parameter sweeps, and main attack conditions; it is not the sample size of one balanced comparison. The readability-priority profile also includes a second capture round for 5 of the 12 content items at each position, giving 459/153 pooled short-exposure/video samples rather than 324/108. Both rounds enter the descriptive means, while duplicate content/position/repeat keys are averaged before paired contrasts. No second long-exposure batch was acquired, so the readability-priority long-exposure row remains $N = 324$. Surya [40], EasyOCR [41], and Tesseract [42] produced 31,725 engine-level records. The main fixed-setting comparison uses the 8 geometries captured at 3.91 ms (Table 4); Table 5 retains the full 9-geometry pool for transparency. The evaluation covers geometrically corrected content crops without an adaptive enhancement sweep. It therefore measures this three-engine, fixed-preprocessing pipeline rather than an upper bound over conventional OCR attacks. Fig. 3 shows digital integrated reconstructions and matched 3.91/31.25 ms captures at the common-setting 1.0 m/0° position for the three primary profiles.

Key findings:

- 1) **Primary fixed-setting evidence:** Table 4 is the main short-exposure estimate. At the common 3.91-ms setting, mean character recovery is 94.5% unprotected, 16.7% for readability-priority, and 5.6% for high-suppression. The corresponding paired content-cluster contrasts are 76.7 percentage points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]). The full 9-geometry sensitivity pool in Table 5 gives 94.1%, 15.1%, and 5.0%, respectively, but includes one 0.49-ms underexposed

TABLE 4. Primary Fixed-Setting Short-Exposure OCR Comparison at the Common 3.91-ms UVC Exposure (Best-of-Engine; Geometrically Corrected Crops; No Adaptive Contrast Enhancement). This Table Excludes the Noncomparable d0.5/15° Underexposed Run and Therefore Defines the Main Short-Exposure Estimate.

Profile	<i>N</i>	Mean char recovery	Paired contrast vs. unprotected	Interpretation
Original (unprotected)	288	94.5%	—	Common-setting baseline
Readability-priority	408	16.7%	76.7 pp [74.3, 79.2]	Conventional-OCR reduction, with upper-tail failures
High-suppression	288	5.6%	88.9 pp [85.5, 91.9]	Exploratory stress profile; misses strict < 5% target

TABLE 5. Single-UVC Real-Capture OCR Sensitivity Pool (Best-of-Engine; *N* = Captures per Row; Pooled Across All 9 Geometric Conditions). Video Rows Average 150 Short-Exposure Frames per Capture Burst. Repeated Content and Geometry Mean That Captures Are Not Independent Population Samples. The Pool Includes the Underexposed d0.5/15° Position; the Primary Short-Exposure Estimate Is Table 4.

Profile	Attack mode	<i>N</i>	Char recovery [95% CI]	Exact match	Sensitive token	Leak rate
Original (unprotected)	Short exposure	324	94.1 [93.1, 95.0]%	65.7%	98.4%	99.4%
	Video (150-frame mean)	108	79.9 [73.5, 86.0]%	61.1%	85.6%	86.1%
	Long exposure	324	47.3 [42.8, 51.9]%	18.8%	95.7%	58.6%
Mask only	Short exposure	324	19.2 [17.0, 21.4]%	0.3%	30.8%	30.6%
	Video (150-frame mean)	108	79.0 [73.0, 84.9]%	48.1%	84.0%	88.0%
	Long exposure	324	67.2 [63.0, 71.5]%	35.5%	94.0%	75.0%
Mask + noise	Short exposure	324	25.6 [22.5, 28.8]%	2.8%	34.4%	36.1%
	Video (150-frame mean)	108	79.2 [73.3, 85.0]%	49.1%	83.9%	88.9%
	Long exposure	324	68.0 [63.7, 72.4]%	39.8%	95.7%	76.2%
Strong (anti-OCR)	Short exposure	324	20.7 [18.2, 23.3]%	0.6%	33.7%	32.4%
	Video (150-frame mean)	108	78.6 [72.5, 84.5]%	50.9%	83.3%	88.9%
	Long exposure	324	61.6 [57.1, 66.1]%	39.8%	95.7%	71.3%
Readability-priority	Short exposure	459	15.1 [13.3, 17.1]%	0.4%	24.0%	20.5%
	Video (150-frame mean)	153	71.1 [65.7, 76.2]%	42.5%	78.5%	85.0%
	Long exposure	324	60.9 [56.6, 65.6]%	36.4%	96.5%	69.1%
High-suppression	Short exposure	324	5.0 [4.3, 5.8]%	0.0%	6.5%	3.7%
	Video (150-frame mean)	108	47.9 [41.8, 54.0]%	5.6%	45.6%	76.9%
	Long exposure	324	9.3 [7.9, 10.9]%	0.0%	34.9%	14.2%

run and is not the primary fixed-setting estimate. The supported positive result is therefore a reduction in this single-camera, fixed-exposure, three-engine pipeline without adaptive contrast enhancement. It is not evidence for an exposure interval or for all conventional-OCR preprocessing strategies.

- 2) **Profile-component and target boundaries:** The mask-only condition reaches 19.2% in the pooled archive, while mask+noise is worse at 25.6% and Strong reaches 20.7%. The archive therefore does not establish a monotonic benefit from noise or weak overlays. Readability-priority is an end-to-end composite observation rather than proof that each component helped. Its pooled sensitive-token recovery is 24.0%, and its P99 character recovery is 94.5%, showing that averages conceal severe failures on some high-value or large-font content. High-suppression also misses its strict < 5% target at the comparable 3.91-ms setting, where its mean is 5.6% [4.8, 6.4]%.
- 3) **Integration attacks remain the residual frontier:** Both long exposure and sustained video temporal averaging exploit full-cycle complementarity (Fig. 4). The readability-priority profile still recovers 60.9% char / 36.4% exact under long exposure (sensitive-token recall 96.5%) and 71.1% char / 42.5% exact under the

150-frame video mean. Excluding the underexposed d0.5/15° position raises readability-priority video recovery to 79.7% (CI [75.4, 83.8]), so the full-pool value understates the practical integration leakage. The divergence between character and sensitive-token recovery reflects the structural properties of high-value fields: digits, account numbers, and URLs are typically short, rendered in large font with a restricted character set, and thus recoverable from degraded images even when surrounding dense text is not. Weak inversion ($\alpha = 0.2$) alone is insufficient to resist temporal integration. The weak Strong overlay likewise provides no measurable integration defense: mask-only and Strong are nearly identical under the 150-frame video mean (79.0% versus 78.6%) and differ only modestly under long exposure (67.2% versus 61.6%), with overlapping bootstrap intervals. Measurable real-capture integration reduction appears only at high-suppression amplitudes, and even there leakage remains substantial.

- 4) **Long-exposure reversal is a notable negative result:** The unprotected long-exposure baseline is only 47.3% char, lower than the readability-priority profile's 60.9%. At the sole 125 ms position (0.5 m/15°), unprotected recovery is 1.1% versus 46.3% readability-priority; the observed clipping of the full-brightness

image supports saturation reduction and a shift into the sensor's approximately linear response region as the explanation for this position. Excluding it, the 31.25 ms anomaly remains position-dependent: at 0.5 m the expected direction holds (unprotected 78–82% > readability-priority 25–55%), whereas several 1.0–1.5 m positions show readability-priority recovery 12–59 pp higher. Without matched local image diagnostics, we only *hypothesize* that residual stripe/glyph edges interact with geometry-specific overexposure or contrast compression; the data do not establish this mechanism. The anomaly does not weaken short-exposure conclusions, where readability-priority is always lower, but it demonstrates that fixed long-exposure parameters produce nonmonotonic effects. The primary validity conclusions are confined to single-frame short exposure; integration attacks are reported as a security boundary.

- 5) **High-suppression profile: reduced integration leakage with unresolved perception cost:** This exploratory stress profile yields 9.3% char (CI [7.9, 10.9]) / 0% exact for long exposure and 47.9% char (CI [41.8, 54.0]) / 5.6% exact (CI [1.9, 10.2]) under the 150-frame video mean. The video leakage is substantial and does not constitute attack closure. The pooled 47.9% masks large inter-position variation: 0.5 m at three angles yields 50.7%/0%/33.8%; 1.0 m yields 61.1–69.4%; 1.5 m yields 48.4–55.5%. The 0% point is the underexposed 0.5 m/15° session discussed in §V-A; excluding it raises video char recovery to 53.9% (CI [48.0, 59.9]) and exact match to 6.3%, confirming that the full-pool result is defense-favorable. Any deployment assessment would need worst-case geometry, exposure, and user-acceptability evidence. The digital pipeline shows lower reconstruction quality for high-suppression, but this is not real-capture or human-subject evidence. The user study protocol (§V-F) excludes this profile, leaving its perceptual cost unquantified.
- 6) **Geometric trends:** Across 9 conditions (0.5–1.5 m × 0–30°), protected short-exposure recovery is consistently below unprotected values. Because the design is unbalanced and no inferential model for geometric factors has been fitted, we do not claim statistically significant monotonic robustness in distance or angle.

Per-engine results: Table 6 breaks down pooled short-exposure results by engine; Fig. 5 visualizes all six profiles. The engines differ widely in unprotected recovery (Surya 37.1%, EasyOCR 79.5%, Tesseract 84.3%). Under Strong, recovery ranges from 2.1–16.3%; under readability-priority, 3.2–11.7%; under high-suppression, 1.8–2.0%. All three show reduction, but the finding is neither recognizer-agnostic nor preprocessing-agnostic. The metric-wise best-of-engine aggregation is attacker-favorable within this limited set, not an upper bound over optimized or commercial OCR pipelines.

Unprotected - Digital integrated reconstruction

The quick brown fox jumps over the lazy dog

Unprotected - Short exposure (single frame)

The quick brown fox jumps over the lazy dog

Unprotected - Long exposure

The quick brown fox jumps over the lazy dog

Readability-priority - Digital integrated reconstruction

The quick brown fox jumps over the lazy dog

Readability-priority - Short exposure (single frame)

The quick brown fox jumps over the lazy dog

Readability-priority - Long exposure

The quick brown fox jumps over the lazy dog

High-suppression - Digital integrated reconstruction

The quick brown fox jumps over the lazy dog

High-suppression - Short exposure (single frame)

The quick brown fox jumps over the lazy dog

High-suppression - Long exposure

The quick brown fox jumps over the lazy dog

FIGURE 3. Real-capture montage at 1.0 m and 0°. Within each vertically stacked profile group (unprotected, readability-priority, and high-suppression), panels show, from top to bottom, the digital integrated reconstruction, 3.91 ms short-exposure capture, and 31.25 ms long-exposure capture. Camera panels are identically cropped from the archived raw camera JPEGs with no geometric correction or brightness gain, so they differ from the geometrically corrected crops used for OCR evaluation. Digital panels use the same 1920 × 1080 black playback canvas and brightness-aligned full-cycle integration for each profile; the readability-priority and high-suppression reconstructions include the recorded $\alpha = 0.2$ inversion slot. These digital views are not direct evidence of human readability or visual comfort.

The evaluated engines are open-source conventional OCR systems; commercial OCR APIs such as Google Cloud Vision, Azure AI Vision, or Baidu OCR were not tested and may achieve higher recovery. The three-engine best-of-engine result should therefore be read as an attacker-favorable aggregation within a limited engine set, not as an upper bound over available OCR services.

C. REAL-CAPTURE VLM PROBES

To test whether conventional OCR conclusions extend to stronger recognizers, we applied three commercial VLM endpoints exposed through the SiliconFlow API (Qwen3-VL-

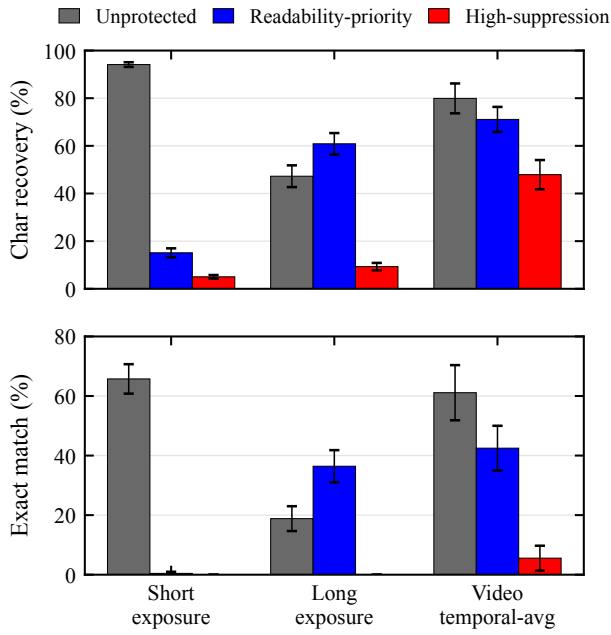


FIGURE 4. Best-of-engine OCR character recovery and exact match for three representative real-capture profiles (unprotected, readability-priority, and high-suppression) under short exposure, long exposure, and a 150-frame video temporal mean, pooled across the 9 tested distance-angle configurations. Lower bars indicate less recovered text. The long-exposure and video bars expose the integration boundary: the readability-priority profile retains substantial recoverable content even when its short-exposure bar is low. Readability-priority sample counts include the disclosed five-item repeat and therefore differ from the other profiles.

TABLE 6. Real-Capture Short-Exposure OCR: Per-Engine Results (Pooled Across 9 Geometric Conditions)

Profile	Engine	Char (%)	Exact (%)	N
Original	Surya	37.1	22.8	324
	EasyOCR	79.5	27.8	324
	Tesseract	84.3	47.8	324
Mask only	Surya	4.2	0.3	324
	EasyOCR	15.6	0.0	324
	Tesseract	3.2	0.0	324
Mask + noise	Surya	11.0	2.8	324
	EasyOCR	18.1	0.0	324
	Tesseract	4.8	0.0	324
Strong	Surya	6.1	0.6	324
	EasyOCR	16.3	0.0	324
	Tesseract	2.1	0.0	324
Readability-priority	Surya	3.2	0.4	459
	EasyOCR	11.7	0.0	459
	Tesseract	3.3	0.0	459
High-suppression	Surya	1.8	0.0	324
	EasyOCR	2.0	0.0	324
	Tesseract	1.9	0.0	324

32B-Instruct [43], Kimi-K2.6, and GLM-4.5V [44]) to the same real-capture photographs. This section is a *boundary probe*: if VLMs recover text where conventional OCR fails, the practical scope of the method is confined to conventional

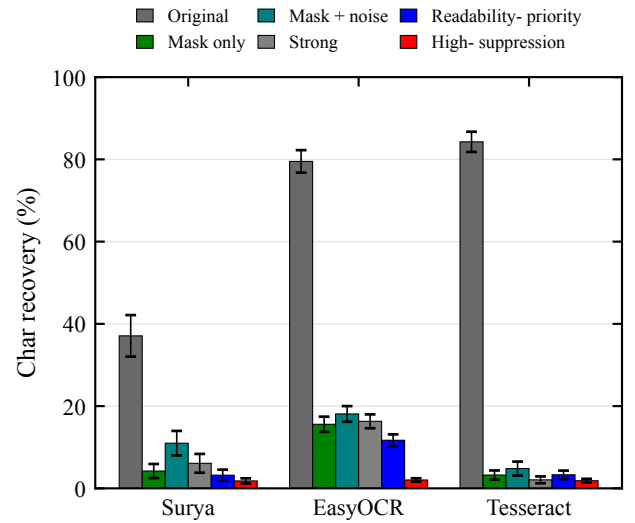


FIGURE 5. Real-capture short-exposure character recovery for three OCR engines across all six profiles in Table 6; error bars are 95% bootstrap intervals resampled by capture. All three engines are suppressed, but with a limited engine family this cannot be claimed as recognizer-agnostic.

OCR short-exposure mitigation.

The probes cover 0° on-axis at three distances. The 0.5 m run includes 156 inputs: 36 unprotected short-exposure base-lines, 36 high-suppression short- and long-exposure inputs each, and four aggregate views from 60 fps video (12 segments each). The 1.0 m and 1.5 m runs each include 120 high-suppression inputs: 36 short- and long-exposure inputs each plus four video aggregate types (12 segments each). VLM inputs and the OCR best-of-engine (BoE) reference in Table 7 use the same file batch (3.91 ms short/video-frame exposure and 31.25 ms long exposure). All reported inputs are raw camera JPEGs without contrast enhancement. Pilot CLAHE-type enhancement partially reconstructed masked text, but no systematic preprocessing sweep was completed. We therefore exclude preprocessing robustness from the supported claim rather than treating raw-JPEG results as an upper bound.

Of 1,188 model requests, 1,133 returned parseable results: 34 failures at 0.5 m (468 calls; Qwen 2, Kimi 0, GLM 32), 21 at 1.0 m (360 calls; Qwen 4, Kimi 1, GLM 16), and none at 1.5 m (360 calls). Each cell reports effective N and is conditional on a parseable response. GLM failures concentrate on mixed-content (47–54%), digit-string (23–38%), and account-credential (31%) items, suggesting content-dependent rather than random missingness. Therefore, cells with failed calls are descriptive and should not be compared directly across models or distances. For the GLM 0.5 m short-exposure cell ($N_{\text{eff}} = 31$), imputing failures as 0% gives 61.0% char / 44.4% exact, while imputing them as 100% gives 74.9% / 58.3%; the reported result is 70.8% / 51.6%.

We use the failure-free 1.5 m run as primary cross-model evidence; the 1.0 m run enters the short-exposure distance sweep as supplementary data. At 1.0 m, Qwen and Kimi

long-exposure exact match are 84.4% and 74.3%, and the successful calls from the three models yield 70.0–83.3% exact match under the 150-frame video mean. GLM long-exposure exact match is only 3.3% at 1.0 m versus 58.3% at 1.5 m, but content-dependent failures preclude a cross-distance consistency claim. For compactness, Table 7 reports only the 150-frame video mean at 0.5 m; the 1.5 m run reports all four aggregate views to show view-selection sensitivity.

All calls used SiliconFlow’s OpenAI-compatible API. The recorded provider identifiers are Qwen/Qwen3-VL-32B-Instruct, Pro/moonshotai/Kimi-K2.6, and zai-org/GLM-4.5V. Qwen3-VL and GLM-4.5V have public technical reports [43], [44]. For Kimi-K2.6, we use the provider endpoint identifier and the archived request metadata as the reproducibility record, rather than making claims about a separately documented upstream public release. The Pro/ prefix in the second identifier is SiliconFlow-specific, and the endpoint accepted image inputs in our archived calls. Decoding temperature is 0 and maximum output is 4,096 tokens. The shared prompt requests only visible text and prohibits guessing; ground truth is not provided. Full prompts, parameters, and per-call outputs are archived. Metrics follow §V-A and are pooled at the capture level. Public CET6 documents may inflate recovery through training-data memorization, so synthetic snippets and exact match provide cleaner discriminative evidence.

0.5 m session selection transparency: The archive preserves an alternative session (real_capture_vlm_d0.5_a0_rerun.json, timestamp 012715) at the same position whose high-suppression short-exposure inputs are nearly all-black, yielding empty transcriptions and 0% char/exact across all three models. Table 7 uses the 141107 (3.91 ms) session because it shares the OCR reference batch and its inputs are not black. These are the only two 0.5 m on-axis VLM sessions, so Qwen’s observed short-exposure exact-match outcome ranges from 0/36 captures in the underexposed session to 28/36 captures in the tabled session, not a stable single-session estimate. The adequate-exposure criterion was applied after inspecting two sessions and is therefore a post-hoc selection rule; preserving both sessions and reporting the range makes the exposure sensitivity explicit. The tabled session is the conservative attacker-favorable case, while the alternative is retained as an underexposure diagnostic.

Table 7 presents the pooled VLM probe results: short-exposure rows show distance variation, while long-exposure and video aggregate rows show integration attack boundaries. Table 8 breaks down 0.5 m high-suppression short-exposure character recovery by content type.

Key findings:

- 1) **VLMs bound rather than extend the claim:** On the same 0.5 m photographs, best-of-engine OCR yields only 8.4% char / 0% exact on the high-suppression short-exposure condition, while Qwen3-VL achieves 91.5% char and exactly recovers 28/36 captures from 12 items. At 1.5 m, the three models retain 33.3–47.2% exact / 61.5–70.0% char. Even with a prompt prohibit-

ing guessing, VLMs recover substantial text from visible fragments. Within this on-axis, high-suppression, three-distance probe, conventional OCR claims therefore do not transfer to VLM attackers; off-axis VLM capture and other protection profiles were not evaluated.

- 2) **Failure concentrates on large-font short snippets:** At 0.5 m, CET6 full-page dense documents yield 0.4–18.5% VLM character recovery (unprotected baseline 64.4%), whereas credentials, code, and digit strings yield 50–100% (Table 8). At 1.5 m, CET6 dense pages drop to 0% char across all three models, while digit strings and short sentences remain highly recoverable. The pattern matches mask granularity: small-font strokes (~ 1 –2 pixels wide) lose glyph topology when $1 - 1/n$ of pixels are removed. Large-font glyphs retain global contours even through sparse fragments, and VLMs can complete them via visual encoding and language priors [45], [46], bypassing the binarization bottleneck that disrupts conventional OCR.
- 3) **Attack capability varies across models and conditions:** Under 0.5 m long exposure, Qwen3-VL recovers 82.4% exact while Kimi/GLM do not exceed 2.8%; at 1.5 m, long-exposure exact rises to 58.3–91.7%. VLM screen-reading ability depends jointly on model version, distance, exposure, and content; any conclusion about “defeating current VLMs” will expire as models update.
- 4) **Sustained video aggregation is the VLM residual frontier:** At 0.5 m, the 150-frame video mean yields 83.3–87.5% exact / $\geq 91.8\%$ char; the three other 0.5 m aggregate views (single-best, window-mean-best, and max-projection) yield Qwen exact of 66.7%, 83.3%, and 83.3% respectively. These views are all derived from the same 150-frame burst, so they characterize view selection after sustained recording rather than independent short-duration attacks. At 1.5 m, the 150-frame mean yields 58.3–66.7% exact / 88.5–90.3% char. By contrast, 1.5 m single-best-frame yields only 0–8.3% exact, and conventional OCR on the same temporal-mean views stays at or below 8.3% exact (48.4–50.7% char). For VLM attackers, multi-cycle integration (§V-E2) is far more dangerous than selecting a single clear frame, but this experiment does not determine the minimum recording duration needed.

The effectiveness claim must therefore be bounded by content and attacker type: the high-suppression profile reduces dense small-text recovery under conventional OCR in this probe, but large-font short snippets (credentials, code, digit strings) remain recoverable when the attacker uses a modern VLM. This negative result repositions the practical scope from “blocking arbitrary machine screen-reading” to “reducing conventional OCR short-exposure batch recovery,” and identifies VLM-aware display protection as an open problem. Countermeasure directions are discussed in §VI-E.

TABLE 7. Real-Capture VLM Probes (0° On-Axis; All Rows Except the Original Short Baseline Are High-Suppression). Each VLM Cell Reports Exact Count / Char / N (Recoveries / % / Effective Calls); Cells with Failures Are Conditional on Parseable Responses. OCR BoE Shows the Same-Batch Three-Engine Best-of-Engine Reference (Exact / Char, %). The Temporal-Mean Rows Average 150 Short-Exposure Frames; Other Aggregate Views Are Defined in the Text. The 0.5 m high-suppression short row uses the adequate-exposure attacker-favorable 141107 session; the only alternative 0.5 m session was underexposed and yielded 0/36 Qwen exact matches.

Distance	Condition	OCR BoE	Qwen3-VL-32B	Kimi-K2.6	GLM-4.5V
<i>Short-exposure distance sweep</i>					
0.5 m	original short	58.3 / 92.4	30 / 94.5 / 36	30 / 95.6 / 36	19 / 92.0 / 26
0.5 m	short	0.0 / 8.4	28 / 91.5 / 36	17 / 84.7 / 36	16 / 70.8 / 31
1.0 m	short	0.0 / 3.6	20 / 79.1 / 36	14 / 68.6 / 36	9 / 47.0 / 33
1.5 m	short	0.0 / 6.5	17 / 68.9 / 36	12 / 61.5 / 36	13 / 70.0 / 36
<i>Key attack views</i>					
0.5 m	long	0.0 / 2.1	28 / 85.7 / 34	1 / 7.9 / 36	0 / 4.9 / 30
0.5 m	Video (150-frame mean)	8.3 / 50.7	10 / 94.0 / 12	10 / 95.5 / 12	7 / 91.8 / 8
1.5 m	long	0.0 / 13.2	33 / 89.3 / 36	32 / 89.9 / 36	21 / 62.4 / 36
1.5 m	Video (single best)	0.0 / 4.3	1 / 24.8 / 12	0 / 12.7 / 12	0 / 29.8 / 12
1.5 m	Video (150-frame mean)	0.0 / 48.4	8 / 90.3 / 12	8 / 90.3 / 12	7 / 88.5 / 12
1.5 m	Video (window-mean best)	0.0 / 8.6	7 / 89.0 / 12	5 / 84.2 / 12	8 / 87.2 / 12
1.5 m	Video (max projection)	0.0 / 19.0	8 / 90.1 / 12	7 / 89.2 / 12	7 / 88.7 / 12

TABLE 8. 0.5 m High-Suppression Short-Exposure VLM Character Recovery by Content Type. Baseline Column Shows Same-Session Unprotected Original Short Results for Qwen (Kimi/GLM Similar); GLM Had Occasional Call Failures, Parenthesized N Shows Effective Count. At 1.5 m, CET6 Dense Pages Yield 0% Char Across All Three Models.

Content type (N)	Baseline (%)	Qwen (%)	Kimi (%)	GLM (%)
CET6 document (3)	64.4	18.5	5.6	0.4
Credentials (6)	97.6	97.6	88.3	77.1 (5)
Code snippets (6)	89.8	94.9	95.8	88.9
Digit strings (6)	100.0	100.0	98.6	50.0 (4)
Mixed content (6)	100.0	100.0	92.5	98.2 (4)
Paragraphs (6)	100.0	99.8	81.7	66.5
English sentences (3)	95.3	95.3	96.9	94.6

The section covers a single camera at 0° on-axis, one profile, three commercial API models, and only 12 video segments per cell; numbers are descriptive boundary probes, not full cross-geometry or cross-model-version evaluation.

D. SUPPLEMENTARY DETECTION AND TRACKING DIAGNOSTICS

Object-detection and tracking diagnostics are reported in the supplementary material rather than the main evidence chain. They use one real-capture position, a limited COCO subset, still-frame MOT captures, and approximate simulation metrics, so they are diagnostic stress tests but do not extend the primary conventional-OCR claim. The normalized cross-task summary figure is omitted from the main paper because OCR character recovery, mAP, higher-order tracking accuracy (HOTA), and identification F1 (IDF1) are not commensurate measurements.

E. SOFTWARE SIMULATION SUPPLEMENTARY EXPERIMENTS

This section uses software simulation to address two questions: (1) what is the machine readability of a single protected subframe obtained from the display compositor?

TABLE 9. Synthetic Single-Subframe OCR: Three-Engine Character Recovery (120 Samples, $n = 4$, $\epsilon = 8/255$)

OCR engine	Original frame	Single subframe	Reduction
Tesseract	94.0%	0.0%	94.0%
EasyOCR	94.1%	0.0%	94.1%
Surya	97.0%	2.6%	94.4%

(2) what is the attacker’s recovery boundary under zero camera noise with perfect alignment? Question (1) also characterizes single-frame screenshots: a screenshot tool captures the framebuffer at one instant, corresponding to one protected subframe. The 0–2.6% recovery rates below apply to this scenario. The boundary is clear: if malware intercepts the original frame before masking, or screen recording accumulates a full cycle, the method offers no protection. We claim only single-frame-level mitigation, not a general “anti-screenshot” guarantee.

1) Synthetic Single-Subframe OCR

Using Tesseract [42], EasyOCR [41], and Surya [40] on 120 synthetic samples, character recovery drops from 94.0–97.0% to 0–2.6% across three engines (Table 9). The result covers the current corpus, parameters, and engine set; stratified breakdowns can check category trends but are insufficient to prove effectiveness across all languages, layouts, or recognizers.

Fig. 6 visualizes these results. Single-subframe recovery does not exceed 2.6% for any engine, confirming the finding is not driven by one recognizer. Broader generalization requires additional engines and corpora.

2) Theoretical Security Boundary

When an attacker under ideal conditions accumulates frames covering a full cycle, the complementarity can be exploited and complementary noise can cancel in the ideal linear model

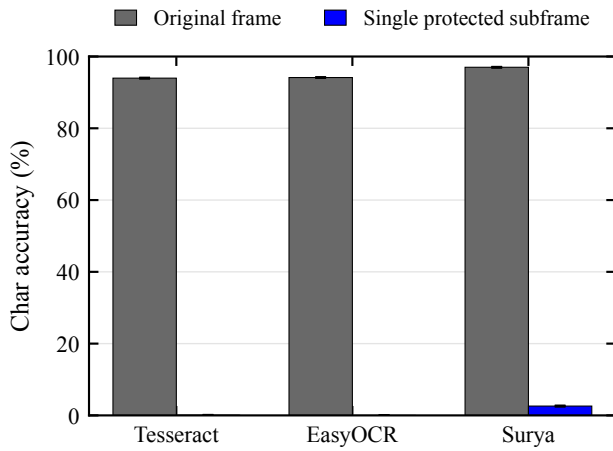


FIGURE 6. Character recovery for three OCR engines on original frames and single protected subframes; error bars are 95% bootstrap intervals resampled by corpus sample.

of (1). The archived digital evaluation recovers 95.2% under the *per-sample strongest attack across the evaluated attack set*; this value is not the pure temporal-mean result. The corresponding pure full-cycle temporal-mean attack recovers 94.3%, with other channel/projection attacks reaching similar values. This is the *fundamental boundary* of an ideal linear temporal decomposition. The high-suppression profile reduces recovery under the current real-capture integration conditions, but the 150-frame video mean still yields 47.9% pooled character recovery and 53.9% after excluding the underexposed position. It can therefore only be called mitigation, not closure of this boundary. Character recovery need not scale with the illuminated-pixel fraction $1/n$, because recognizer performance also depends on glyph topology, image content, display response, optics, and ISP processing. The current experiments do not provide a formal information-theoretic bound.

Additional software-proxy baseline, parameter-sweep, detection, and tracking diagnostics are preserved outside the main evidence chain. They are not used as main-text evidence because their physical conditions, user-experience costs, and metrics are not matched to the real-capture OCR estimand.

F. USER EXPERIENCE STUDY

To evaluate the impact of the protected display on human readability and visual comfort, we designed a web-based user study.

Equipment and environment protocol: The same 240 Hz display as in §V-A. The formal study version sets a hard measured refresh-rate threshold of ≥ 200 Hz to ensure the fixed $n = 4$ basic subframe cycle rate $f_r/n \geq 50$ Hz; below this threshold participation is blocked rather than silently reducing n . The 144–199 Hz adaptive path is available only for an explicitly labeled demonstration mode, and its sessions are excluded from analysis by default. Before each participant begins, the experimenter confirms fixed display brightness,

disabled auto-brightness/power-saving and variable refresh rate, unified browser fullscreen, closed high-load background processes, and an approximate 60 cm viewing distance. Refresh rate is re-measured after any display-mode change.

The browser records actual frame intervals in each trial's `requestAnimationFrame` loop; if the interval between adjacent callbacks exceeds $1.5\times$ the expected interval computed from the pre-trial measured refresh rate, a dropped-frame event is logged, along with estimated drop count, drop rate, mean/max frame intervals, observed refresh rate, basic subframe cycle rate, and full cycle rate with inversion frame. This recording reflects only browser presentation cadence and cannot substitute for panel scanning or photometric measurement.

Ethics and safety: The planned study involves minimal-risk human-subjects research, and no participant data have been collected. Before recruitment, the authors will obtain and document a written approval, exemption, or waiver determination from the competent institutional or departmental authority; formal data collection will not begin without that determination. Participants must separately check informed consent and photosensitivity screening: the page explicitly states that the masked display involves rapid temporal modulation, that participants should stop immediately upon experiencing discomfort, eye fatigue, dizziness, nausea, or headache, and that they may withdraw unconditionally at any time. Individuals who self-report a history of photosensitive epilepsy or sensitivity to flicker are not permitted to proceed. The system records consent timestamps and screening pass flags. The implementation applies a conservative $f_r/n \geq 50$ Hz operating floor, but this engineering gate is not an ethical or clinical safety determination. At $n = 4$ on a 240 Hz panel the basic cycle rate is 60 Hz; the inversion frame reduces the full cycle frequency to $f_r/(n + 1) = 48$ Hz, and the implementation separately logs warnings and actual timing. Registration information (student ID, name) is used only for participation management; analysis and publication use only de-identified data. Age and gender are optional demographic fields.

Task design: Each participant first completes one 12-second, unscored unmasked Canvas typing warm-up, views a 10-second, unscored readability-priority full-mask preview, and then completes an 8-second, unscored typing practice under the same readability-priority mask. This sequence separates initial stimulus and input adaptation from the four subsequent 20-second scored trials (within-subjects design). On every typing screen, including warm-up, practice, and scored trials, the stimulus Canvas is visible and its player has started before the participant clicks “Start.” Clicking “Start” triggers a 5-second countdown while the input remains disabled; the timed typing interval and input availability begin only when the countdown ends. The same pre-start viewing procedure applies to both control and masked conditions.

The two conditions are each repeated twice. The experimenter's continuously assigned zero-based sequence number k selects ABBA or BAAB order to counterbalance practice

and fatigue: (A) **original text** (control), displayed as unmasked Canvas with $n = 1$; (B) **masked text**, using the readability-priority configuration: $n = 4$, mask+noise, strong anti-OCR ($\alpha_s = 0.10$, $\alpha_g = 0.12$, 6 cycles) + weak inversion ($\alpha = 0.2$). The typing order index is $k \bmod 2$, and the subjective-rating Latin-square row is $\lfloor k/2 \rfloor \bmod 6$. Every consecutive 12 participants therefore cover all 2×6 joint assignments. Formal sessions enforce a uniqueness constraint on k , and the backend recomputes and verifies both order indices.

Both conditions are rendered by the same `render SourceCanvas` path, with identical dark background, light text, canvas size, 23 px bold font, and color. The sole processing difference is the temporal protection pipeline, avoiding polarity and font-weight confounds. Here “cycles” means that the web implementation rotates through six distinct mask, noise, and stripe instances before repeating. This web-playback-specific setting shares α_s and α_g with the real-capture configuration in §IV and avoids repeatedly integrating one fixed overlay pattern.

Scoring no longer compares characters position-by-position but aligns the full transcription against the optimal target prefix using Levenshtein minimum string distance (MSD) [47]; untyped target suffix does not count as error. The alignment yields accuracy, characters per minute (CPM), and words per minute (WPM), along with edit distance, aligned prefix length, and scoring version. First-keystroke latency is defined from the end of the 5-second countdown, when the input is enabled and the scored timer starts, to the first input event that makes the transcription non-empty. Analysis first averages each participant’s two trials per condition, then performs within-subject paired comparison.

Subjective ratings: After the typing trials, participants rate 6 conditions on the three 1–5 Likert items displayed by the interface: readability (1 = difficult to read, 5 = very clear), stability (stored in the `flicker` field; 1 = strong flicker, 5 = no perceptible flicker), and immediate visual comfort (`fatigue`; 1 = very uncomfortable, 5 = very comfortable). Higher values therefore indicate a better outcome on every item. Conditions include the unmasked original anchor, $n \in \{2, 3, 4\}$ mask+noise, $n = 4$ mask-only, and the same $n = 4$ readability-priority full configuration (mask+noise+strong anti-OCR+weak inversion) used in the typing task. The first three masking levels all achieve ≥ 60 Hz real temporal playback on a 240 Hz panel; the originally planned $n = 8$ would yield 30 Hz and trigger static fallback, so it is not included as a temporal ablation. The six conditions are presented in a 6th-order balanced Latin square; each condition must be viewed for at least 10 seconds before submission is enabled, with viewing start/end times and duration recorded. The unmasked anchor is also an attention/scale-interpretation aid but is not mechanically excluded by a “must rate 5” rule. This study does not include the high-suppression profile; assessment of its visible stripes and grain’s acceptability is left to future work (see §VI).

Sample size and analysis plan: A minimum of $N = 24$

TABLE 10. Typing Task: Within-Subject Comparison of the Control and Readability-Priority Masked Conditions ($N = XX$; Participant Means Over Two Trials per Condition; ‘ = Masked – Control With Participant-Level Bootstrap 95% CI). Per Metric, a Paired t -Test Is Used When the Pre-Specified Shapiro Test Passes (Effect Size Cohen’s d_z), Otherwise a Wilcoxon Signed-Rank Test (Effect Size Rank-Biserial r_{rb}); Test Statistics Are Reported in the Text

Metric	Control	Masked	Δ [95% CI]
Accuracy (%)	–	–	– [–, –]
WPM	–	–	– [–, –]
CPM	–	–	– [–, –]
Attempted chars	–	–	– [–, –]
First-key latency (ms)	–	–	– [–, –]

participants is planned, a number within the range required for paired effect sizes of $d_z \approx 0.6$ – 0.8 at 80% power, and exactly covering two rounds of the 12-person joint order assignment. Pre-registered exclusion criteria are debug/demo sessions, incomplete four typing or six rating records, measured refresh rate below 200 Hz, any typing trial with fewer than 5 attempted characters, control mean accuracy below 50%, masked trials not in temporal mode, any trial with observed basic cycle rate below 50 Hz, or minimum-view straight-line ratings. The last criterion is defined as all six rating rows having a recorded viewing duration no greater than 11 seconds (a 1-second tolerance above the 10-second submission gate) and, within every row, identical scores on all three items. WPM, CPM, accuracy, attempted characters, and first-keystroke latency are first averaged within each participant by condition; speed metrics must be interpreted jointly with accuracy to avoid misinterpreting fast input under low accuracy as performance improvement. Paired differences are tested using paired t -tests when the pre-specified Shapiro test is passed, otherwise Wilcoxon signed-rank tests, with effect sizes and participant-level bootstrap 95% confidence intervals reported. Likert data are analyzed per dimension using Friedman tests with Holm-corrected post-hoc pairwise Wilcoxon tests. Analysis scripts generate de-identified participant means, JSON audit reports, and two $\LaTeX$ tables directly from the formal database `study_formal.db`.

Participants: XX volunteers were recruited; XX sessions met all pre-registered inclusion criteria and entered analysis (XX excluded; per-criterion audit counts accompany the released analysis output). Optional demographics for the included sample: XX of XX reported age (median XX years, range XX–XX) and XX reported gender (XX women, XX men). All formal sessions ran on the 240 Hz laboratory display with a measured refresh rate of at least 200 Hz.

Results: Table 10 and Fig. 7 summarize the typing task. Relative to the control condition, the readability-priority full configuration changed mean accuracy by XX percentage points (95% CI [XX, XX], $p = XX$, effect size XX), WPM by XX (95% CI [XX, XX], $p = XX$, effect size XX), and first-keystroke latency by XX ms (95% CI [XX, XX], $p = XX$, effect size XX); CPM and attempted characters gave $p = XX$ and XX, respectively. Table 11 and Fig. 8 summarize the

TABLE 11. Subjective Ratings for the Six Display Conditions ($N = XX$; 1–5 Likert, Higher Is Better; Mean $\pm$ SD). Stability Is the Inverse-Flicker Item and Comfort the Immediate Visual Comfort Item (§V-F); Friedman and Holm-Corrected Pairwise Wilcoxon Statistics Are Reported in the Text

Condition	Readability	Stability	Comfort
Unmasked anchor	–	–	–
$n=2$ mask+noise	–	–	–
$n=3$ mask+noise	–	–	–
$n=4$ mask+noise	–	–	–
$n=4$ mask-only	–	–	–
Readability-priority full ($n=4$)	–	–	–

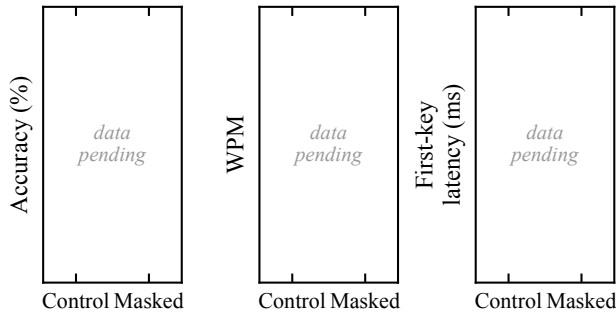


FIGURE 7. Within-subject typing performance under the control and readability-priority masked conditions: each gray line is one participant (mean of two trials per condition); markers show group means with participant-level bootstrap 95% confidence intervals. Panels show transcription accuracy, words per minute, and first-keystroke latency.

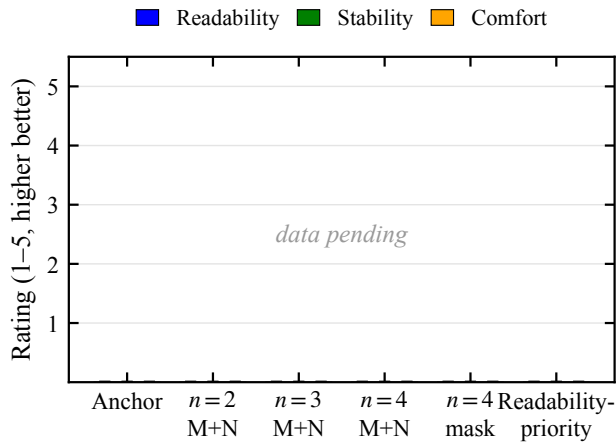


FIGURE 8. Mean 1–5 Likert ratings (error bars: SD) for readability, stability, and immediate visual comfort across the six rated display conditions, ordered from the unmasked anchor to the readability-priority full configuration. Higher is better on every item.

subjective ratings: Friedman tests yielded $\chi^2 = XX$ ($p = XX$) for readability, $\chi^2 = XX$ ($p = XX$) for stability, and $\chi^2 = XX$ ($p = XX$) for immediate visual comfort; Holm-corrected pairwise comparisons between the readability-priority full configuration and the unmasked anchor gave $p = XX$, XX , and XX on the three items, respectively.

VI. DISCUSSION

A. EVIDENCE HIERARCHY AND CLAIM SCOPE

The most directly supported conclusion is a fixed-condition observation. At the common 3.91-ms setting on one S600 link, the evaluated profiles reduce mean recovery for three open-source OCR engines applied to geometrically corrected crops without adaptive enhancement. This is not an exposure interval, a general anti-photography claim, or evidence against an optimized attacker. The 10,575-image archive reflects component, parameter, and attack-condition coverage rather than the sample size of one balanced comparison. The unbalanced corpus, single webcam, missing luminance-matched controls, and fixed preprocessing prevent cross-device, cross-scenario, or mechanism-isolation inference. VLM, integration, simulation, and supplementary detection/tracking results define boundaries rather than extend the primary claim. The passive baselines are software proxies, so the data also do not establish superiority over physical filters, prior temporal-display systems, or differently tuned static obfuscation.

B. THE MATHEMATICAL TENSION OF LINEAR INTEGRATION

The scheme exploits a difference between human temporal perception and a short camera exposure. Its mathematical boundary, however, is defined in an ideal linear display–camera radiance model rather than by treating human vision as a linear summation operator. In that model, complementary subframes can be re-summed to recover the original ($\sum I_k = I$), so a camera that accumulates a registered full cycle defeats the linear decomposition. Research on VLM adversarial robustness [46] further shows that large models extract partial semantic information from perturbed images; the probes in §V-C confirm this. VLMs recover large-font content from single short-exposure photographs and obtain stronger signals from the 150-frame video mean and long-exposure views. The archived digital evaluation reaches 95.2% under the per-sample strongest attack across all evaluated reconstructions; pure full-cycle temporal-mean recovery reaches 94.3% (§V-E2).

C. PHYSICAL-LAYER CONFOUNDS AND THE SIMULATION-TO-REAL GAP

Two physical-layer mechanisms systematically separate real-capture results from idealized single-subframe simulation.

Direction 1: real short-exposure recovery exceeds simulation (favoring the attacker). Simulated single-subframe OCR yields 0–2.6% character recovery (Table 9), whereas real-capture short-exposure recovery under comparable masking is 19.2% (mask-only, Table 5). Two physical effects could contribute. First, rolling-shutter row mixing can make different rows sample different display phases, although the S600 readout time and phase were not measured. Second, LCD response transitions can blend adjacent subframes; the AOC 24G51Z’s specified 1 ms GtG already occupies about 24% of a 4.17 ms slot. The real-capture measurements include both effects, but this experiment does not identify their sep-

arate contributions. The simulation is therefore an idealized single-subframe reference, not a physical lower bound.

Direction 2: real integration recovery is lower than simulation (favoring the defender). The per-sample strongest digital attack recovers 95.2% (§V-E2), whereas the real-capture 150-frame video mean yields 71.1% for the readability-priority profile and 47.9% for high-suppression in the full pool. After the underexposed position is excluded, these increase to 79.7% and 53.9%, respectively. Camera ISP nonlinearities (gamma, white balance, noise reduction), moiré interference between the display pixel grid and the sensor Bayer pattern, optical blur from defocus and lens aberration, and imperfect temporal registration across frames all degrade the attacker’s integration quality. Even the more conservative sensitivity values remain below the digital strongest-attack result, but the gap is smaller than the pooled figures suggest; a purpose-built attack setup could narrow it further.

These two directions imply that the simulation and real-capture experiments answer different questions: simulated single-subframe recovery characterizes the algorithm’s theoretical information hiding, while real-capture recovery characterizes the end-to-end system including panel, camera, and ISP confounds.

D. ROLE AND COST OF THE EXPLORATORY HIGH-SUPPRESSION PROFILE

The exploratory high-suppression profile breaks strict completeness through stripe and glyph perturbations, reducing integration recovery at the cost of worsened reconstruction metrics. In the digital pipeline, its full-cycle SSIM is 0.764 versus 0.948 for readability-priority, and its ΔE_{00} is 4.87 versus 1.46, i.e., substantially worse reconstruction quality on both metrics. However, the same digital ablation exposes a direct single-slot failure: inverting and contrast-stretching the weak inversion frame recovers 93.0% character accuracy on average and yields 81.2% exact match. This attack also determines the profile’s 93.0% best-observed digital recovery, so the lower 56.6% full-cycle temporal-average result is not a worst-case security bound. In the full real-capture pool it achieves 0% short-exposure exact match and 5.6% exact match under the 150-frame video mean, while video character recovery remains 47.9%; after excluding the underexposed position, short-exposure character recovery is 5.6% and video character recovery is 53.9% (§V-B). It therefore misses the $< 5\%$ short-exposure target and only mitigates integration leakage. Whether users accept the visible stripes and grain remains untested: the study in §V-F covers only the readability-priority profile and core mask conditions.

This profile split is central to interpretation. The readability-priority profile is the only profile connected to the planned readability and comfort study, but it is also the weaker protection setting under sensitive-token and integration attacks. The high-suppression profile lowers full-cycle temporal-average and real-capture recovery, but its directly exploitable inversion slot prevents treating it as a generally secure setting; it likewise lacks human-subject acceptability

evidence. The present manuscript therefore supports a feasibility niche and an evidence hierarchy, not a completed deployment recommendation.

E. VLM ATTACKERS AND COUNTERMEASURE DIRECTIONS

Protection failure concentrates on large-font short snippets (§V-C), and temporal averaging further amplifies the signal: glyph contour fragments in a single subframe suffice for language-prior completion, and multi-cycle coverage approaches the original. We retain this negative result as boundary evidence because it directly constrains the method’s deployability: the scheme mitigates conventional OCR short-exposure capture but does not resist modern VLM screen-reading. Three unvalidated research hypotheses follow: (1) **font-size-adaptive mask granularity**, scaling the mask cell with stroke width so that a single subframe no longer preserves character topology; (2) **glyph decoys**, testing whether false strokes can reduce single-frame VLM completion; and (3) **field-level content policies**, testing dense small-font rendering or exploratory high-suppression variants on high-risk fields. The decoy idea cannot be presented as a solution within the current $\sum_k \mathbf{N}_k = \mathbf{0}$ framework: any exactly complementary decoy also cancels under full-cycle integration and inherits the same linear boundary. None of these directions has been experimentally validated, and adversarial perturbations targeting VLM encoders are further constrained by digital-to-physical transferability (§II). More generally, any information visible to the observer within the integration window is in principle recoverable by a full-cycle camera paired with a strong-prior model; VLMs merely reduce the fragment count needed. The defense objective should shift from “unreadable to any attacker” toward increasing attack cost and reducing unattended bulk collection.

F. SUSTAINED VIDEO ATTACKER

The evaluated video attack does not establish a four-frame by-pass. The capture pipeline records 150 short-exposure frames at nominal 60 fps, and the reported temporal-mean image averages the complete burst, corresponding to approximately 2.5 s of nominal capture. This attack does not require the mask seed, phase, or precise frame selection because sustained averaging accumulates multiple display cycles. Its pooled recovery is 71.1% for the readability-priority profile and 47.9% for high-suppression, rising to 79.7% and 53.9% after excluding the underexposed position. The archive also contains aggregate views selected from the same full burst, but they do not establish recovery from an independently acquired short window. A duration-versus-recovery curve and pre-specified short-window analysis remain unevaluated. The single-shot scope should therefore be distinguished from the measured sustained-recording boundary without assigning an unsupported minimum bypass duration.

G. DEPLOYMENT PATH

Real-world deployment would require OS driver-level framebuffer injection (DXGI / KMS-DRM / CoreDisplay),

panel/backlight timing coordination, and high-refresh panels meeting the target cycle rate. These steps introduce color management, luminance saturation, frame drops, and synchronization errors that could alter algorithm behavior; they cannot be treated as engineering packaging that leaves conclusions unchanged. The readability-priority profile's 96.5% sensitive-token recovery under long exposure means credentials and similar high-risk fields should not rely on it alone. Candidate field-level mitigations, such as exploratory high-suppression variants or dense small-font rendering, would require separate readability, exposure, video, and VLM evaluation before they could be recommended.

H. ETHICS STATEMENT

This research is defensive in nature, aiming to protect screen content from unauthorized photography. The attack analysis (§V-E2) is intended for honest assessment of security boundaries, not as an attack guide. The planned user study would involve human subjects in minimal-risk research; no participant data have been collected. Recruitment will begin only after the competent institutional or departmental authority issues a written approval, exemption, or waiver determination. The protocol specifies informed consent, photosensitivity screening, and voluntary withdrawal as described in §V-F.

I. LIMITATIONS

- 1) The user study (§V-F) is a protocol, not completed evidence: no participant data exist yet, so the readability-priority profile's usability remains unvalidated. Even when conducted in a controlled 240 Hz laboratory, external validity will require broader participant samples and display conditions.
- 2) Immediate visual comfort is a single-item scale measured after short exposure and does not extend to long-term fatigue or clinical visual discomfort.
- 3) The PoC is application-layer only, without OS drivers or hardware backlight. Luminance, HDR, and color-difference conclusions come from digital models, lacking photometric or panel-response measurements. Actual playback slot timing was not verified with a photodiode, high-speed camera, or oscilloscope. OS compositor jitter, VSync mismatch, and dropped frames can shift subframe boundaries unpredictably; if a short-exposure window straddles the intended slot edge, the camera captures cross-slot content and real recovery may exceed the values reported here. Because the experiment lacks a luminance-matched static control and matched noise-off enhanced profiles, it cannot separate temporal sparsity, duty-cycle luminance, static overlays, panel response, camera exposure, and ISP processing as mechanisms.
- 4) Real-capture OCR used one S600 UVC webcam; ambient illuminance was not recorded; exposure values have two calibration sets. The unbalanced design with repeated content means bootstrap intervals do not substitute for cross-device replication. Uncontrolled ambi-

ent light variation may affect results through two paths: (a) the camera's auto-exposure, auto-white-balance, and auto-gain algorithms respond to ambient conditions, altering effective exposure on the screen content; (b) ambient light reflected off the screen surface changes the captured signal-to-noise ratio. If geometric configurations were captured at different times of day, systematic ambient variation could confound geometric-factor analysis. Increased reflection or sensor noise would most likely suppress the already weak protected signal more than the unprotected baseline, biasing protected recovery downward and favoring the defense; automatic gain or exposure compensation could act in the opposite direction. Because illuminance was not logged, the net direction and magnitude cannot be estimated. This uncertainty primarily weakens cross-geometry comparisons; the large within-link short-exposure reduction remains directly observed but is not immune to the confound.

- 5) Real-capture detection used 150 COCO images at a single position; tracking used 450 still-frame captures from one MOT17 sequence, not continuous video. Simulated detection used 8 images with approximate metrics.
- 6) VLM evaluation covers one camera, 0° on-axis, one profile, and three commercial API models (12 video segments per cell). Models may update over time. Public document content may inflate recovery via training-data memorization. We did not evaluate multi-cycle adaptive reconstruction, rolling-shutter phase search, or pre-masking malware interception. Full-cycle integration remains the fundamental boundary.
- 7) Conventional OCR evaluation covers Surya, EasyOCR, and Tesseract but no commercial OCR API. Best-of-engine over these three systems is attacker-favorable within the evaluated set, yet Google Cloud Vision, Azure AI Vision, Baidu OCR, or future recognizers may obtain higher recovery.
- 8) Baseline comparisons use unmatched software proxies rather than physically tested privacy filters or prior temporal-display systems. They contextualize the observed trade-off but do not establish comparative superiority over those alternatives.
- 9) The readability-priority profile is not suitable as the sole protection layer for sensitive fields: complete sensitive-token recovery remains 24.0% under short exposure and 96.5% under long exposure. Field-level masking, independent access controls, and exposure to the video/VLM boundary must be considered together.
- 10) All 12 test content items are English/ASCII text. CJK characters (Chinese, Japanese, Korean) have much higher stroke density than Latin letters; at the same mask granularity, CJK strokes are more likely to be disrupted (stroke widths closer to mask cell size), but higher inter-character visual similarity may also increase OCR confusion. The net effect on protection

efficacy for non-Latin scripts is unknown and left to future work.

VII. CONCLUSION

We measured profile-level temporal display processing at one fixed exposure on a single UVC link. Across the 8 geometries acquired at 3.91 ms, mean best-of-engine character recovery was 94.5% unprotected, 16.7% for readability-priority, and 5.6% for high-suppression. Paired content-cluster contrasts were 76.7 percentage points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]), respectively. These results support a reduction for three open-source OCR engines under fixed geometric correction and preprocessing. They do not establish an exposure interval or robustness to adaptive enhancement. Readability-priority also retains 24.0% pooled sensitive-token recovery and 94.5% P99 character recovery, while high-suppression misses its strict $< 5\%$ target at the common setting.

The bypass evidence is equally restrictive. A 150-frame temporal mean recovers 71.1% character for readability-priority and 47.9% for high-suppression; the common-setting values are 79.7% and 53.9%. In the full 9-geometry sensitivity pool, long-exposure recovery is 60.9% for readability-priority versus 47.3% unprotected. On-axis VLM probes also recover substantial large-font content. Sustained recording, long exposure, adaptive preprocessing, and modern VLMs therefore remain outside the supported condition.

The study therefore contributes a bounded physical measurement and a set of failure constraints, not a deployable anti-camera defense. The next decisive experiments are an exposure–recovery curve, a systematic preprocessing attack sweep, usability testing of stronger suppression, luminance-matched controls, and cross-device replication.

DATA AND CODE AVAILABILITY

The software, experimental configurations, de-identified per-sample OCR results, VLM probe records, and figure-generation scripts are maintained in the project repository at <https://github.com/docandyc/privacy-display>. An immutable release aligned with the final submitted manuscript, figures, analysis scripts, and de-identified records will be minted before or with the submission package; no release identifier is claimed here. VLM records include provider endpoint identifiers, request timestamps, prompts, decoding parameters, input-image hashes or paths, parse status, and raw model outputs. These fields are required because Kimi-K2.6 is cited through the provider endpoint and archived call metadata rather than an independent public technical report. The canonical archived identifier `capture_hardened` and the legacy `v1m` tag both denote the paper's high-suppression configuration (§IV-E). If the user study is conducted, its database `study_formal.db` (containing student IDs and names) will not be released; public materials will include only de-identified results, exclusion audits, and analysis scripts.

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[PLACEHOLDER: Acknowledgment to be filled.]

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